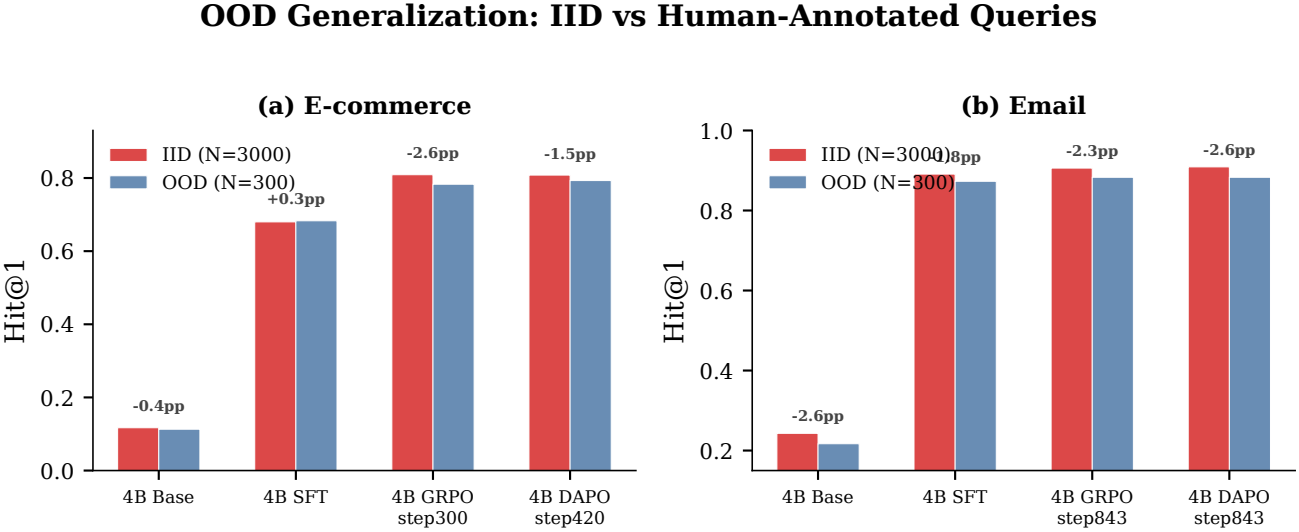
OOD Generalization: IID vs Human-Annotated Queries
(a) E-commerce
IID (N=3000)
OOD (N=300)
Hit@1
0.0
0.2
0.4
0.6
0.8
-0.4pp
+0.3pp
-2.6pp
-1.5pp
4B Base
4B SFT
4B GRPO step300
4B DAPO step420
(b) Email
IID (N=3000)
OOD (N=300)
Hit@1
1.0
0.8
0.6
0.4
0.2
-2.6pp
-2.3pp
-2.6pp
4B Base
4B SFT
4B GRPO step843
4B DAPO step843